

## HA TINH, Viet Nam

WMO: 488460

Lat: 18.350N

Lon: 105.900E

Elev: 3

StdP: 101.29

Time Zone: 7.00 (E07)

Period: 96-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 11.5       | 12.4 | 8.1                           | 6.7 | 13.6 | 9.5 | 7.4 | 13.9 | 5.3                   | 16.7 | 4.5 | 17.4 | 1.6                   | N/A  | 0.260 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 6             | 5.7                    | 36.2            | 27.5 | 35.3 | 27.3 | 34.4 | 27.1 | 28.6                | 33.8 | 28.0 | 33.3 | 27.6 | 32.7 | 2.3                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.2                            | 23.0 | 31.4 | 26.7 | 22.3 | 30.8 | 26.3 | 21.8 | 30.3 | 92.7          | 34.1 | 90.2 | 33.6 | 88.2 | 32.8 | 31.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |     |            |     |            |     |      |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-----|------------|-----|------------|-----|------|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |     | n=20 years |     | n=50 years |     |      |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max | Min        | Max | Min        | Max |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l) | (m)        | (n) | (o)        | (p) |      |  |
| 5.0               | 4.2  | 3.5 | DB                         | 10.6 | 38.2               | 0.7 | 0.9                                                | 10.1 | 38.9       | 9.7 | 39.5       | 9.3 | 40.0       | 8.7 | 40.7 |  |
|                   |      |     | WB                         | 9.2  | 29.6               | 1.2 | 0.8                                                | 8.3  | 30.2       | 7.6 | 30.7       | 6.9 | 31.1       | 6.0 | 31.7 |  |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 24.7   | 17.7  | 19.0  | 21.2  | 25.4  | 28.5  | 30.5  | 30.3  | 29.3  | 27.5  | 25.3  | 22.6  | 19.0  |
|                                                                   | DBStd       | 5.21   | 3.10  | 3.73  | 3.50  | 3.17  | 2.73  | 2.08  | 1.68  | 1.54  | 1.78  | 1.98  | 2.74  | 2.79  |
|                                                                   | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                   | HDD18.3     | 126    | 52    | 34    | 13    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 25    |
|                                                                   | CDD10.0     | 5374   | 238   | 252   | 347   | 461   | 574   | 615   | 631   | 599   | 526   | 474   | 377   | 279   |
|                                                                   | CDD18.3     | 2458   | 31    | 53    | 102   | 211   | 316   | 365   | 372   | 341   | 276   | 216   | 130   | 46    |
|                                                                   | CDH23.3     | 26818  | 42    | 146   | 481   | 1936  | 3936  | 5153  | 5203  | 4433  | 3044  | 1713  | 647   | 87    |
| Wind                                                              | WSAvg       | 1.3    | 1.3   | 1.0   | 1.1   | 1.1   | 1.3   | 1.4   | 1.3   | 1.3   | 1.3   | 1.6   | 1.6   | 1.5   |
|                                                                   | PrecAvg     | 2400   | 65    | 46    | 62    | 72    | 181   | 118   | 113   | 244   | 474   | 628   | 210   | 124   |
| Precipitation                                                     | PrecMax     | 3356   | 144   | 84    | 144   | 188   | 469   | 399   | 302   | 666   | 1007  | 1301  | 482   | 380   |
|                                                                   | PrecMin     | 1605   | 0     | 8     | 12    | 11    | 52    | 8     | 27    | 47    | 153   | 127   | 25    | 7     |
|                                                                   | PrecStd     | 445    | 38    | 23    | 31    | 44    | 96    | 96    | 72    | 147   | 217   | 329   | 126   | 83    |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 25.9  | 29.1  | 33.2  | 36.5  | 37.2  | 37.2  | 37.3  | 36.1  | 34.2  | 31.5  | 29.6  | 27.0  |
|                                                                   |             | MCWB   | 22.7  | 24.4  | 24.4  | 27.5  | 27.9  | 27.4  | 27.6  | 27.3  | 27.4  | 26.2  | 25.5  | 24.3  |
|                                                                   | 2%          | DB     | 24.4  | 26.1  | 29.0  | 33.7  | 35.7  | 36.0  | 35.7  | 34.5  | 32.9  | 30.2  | 28.6  | 25.4  |
|                                                                   |             | MCWB   | 21.8  | 23.1  | 24.5  | 27.0  | 27.7  | 27.2  | 27.5  | 27.0  | 26.8  | 25.5  | 24.8  | 23.0  |
|                                                                   | 5%          | DB     | 23.3  | 24.9  | 27.0  | 31.8  | 34.5  | 35.1  | 34.8  | 33.5  | 31.8  | 29.3  | 27.6  | 24.3  |
|                                                                   |             | MCWB   | 21.1  | 22.6  | 24.0  | 26.4  | 27.4  | 27.0  | 27.3  | 26.7  | 26.4  | 25.1  | 24.3  | 22.1  |
|                                                                   | 10%         | DB     | 22.2  | 24.0  | 25.6  | 30.0  | 33.1  | 34.2  | 33.8  | 32.6  | 30.8  | 28.5  | 26.6  | 23.2  |
|                                                                   |             | MCWB   | 20.4  | 22.0  | 23.3  | 25.8  | 27.0  | 26.8  | 26.9  | 26.4  | 26.1  | 24.8  | 23.8  | 21.2  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 23.3  | 24.7  | 25.7  | 28.6  | 29.6  | 28.7  | 28.8  | 28.7  | 28.1  | 27.2  | 26.2  | 24.9  |
|                                                                   |             | MCDB   | 25.0  | 28.0  | 29.6  | 34.0  | 35.3  | 34.0  | 34.0  | 33.1  | 32.4  | 30.3  | 28.7  | 26.3  |
|                                                                   | 2%          | WB     | 22.4  | 23.6  | 25.0  | 27.5  | 28.5  | 28.1  | 28.2  | 27.7  | 27.4  | 26.3  | 25.2  | 23.4  |
|                                                                   |             | MCDB   | 23.9  | 25.5  | 28.0  | 32.2  | 34.0  | 33.7  | 33.7  | 32.0  | 31.3  | 28.8  | 27.6  | 25.0  |
|                                                                   | 5%          | WB     | 21.4  | 22.8  | 24.3  | 26.7  | 27.8  | 27.6  | 27.7  | 27.3  | 26.9  | 25.7  | 24.6  | 22.4  |
|                                                                   |             | MCDB   | 22.9  | 24.6  | 26.8  | 30.6  | 32.8  | 33.3  | 33.1  | 31.6  | 30.4  | 28.0  | 26.9  | 23.9  |
|                                                                   | 10%         | WB     | 20.5  | 22.2  | 23.5  | 26.1  | 27.2  | 27.2  | 27.4  | 26.9  | 26.5  | 25.3  | 24.1  | 21.4  |
|                                                                   |             | MCDB   | 22.0  | 23.9  | 25.5  | 29.6  | 31.9  | 32.8  | 32.6  | 31.2  | 29.7  | 27.5  | 26.3  | 23.0  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 3.3   | 3.3   | 3.9   | 4.7   | 5.6   | 5.7   | 5.6   | 5.5   | 4.7   | 4.1   | 3.9   | 3.6   |
|                                                                   |             | MCDBR  | 4.8   | 4.6   | 6.1   | 8.1   | 7.5   | 7.1   | 7.2   | 7.0   | 6.5   | 5.5   | 5.2   | 4.6   |
|                                                                   |             | MCWBR  | 2.5   | 2.4   | 2.6   | 3.4   | 2.9   | 2.6   | 2.6   | 2.6   | 2.3   | 2.2   | 2.3   | 2.5   |
|                                                                   | 5% WB       | MCDBR  | 4.3   | 4.6   | 5.7   | 6.7   | 6.6   | 6.3   | 6.3   | 6.2   | 5.8   | 4.8   | 4.8   | 4.2   |
|                                                                   |             | MCWBR  | 2.6   | 2.6   | 2.8   | 3.2   | 3.1   | 2.9   | 2.8   | 2.7   | 2.5   | 2.2   | 2.7   | 2.6   |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.578 | 0.611 | 0.717 | 0.667 | 0.526 | 0.469 | 0.448 | 0.483 | 0.553 | 0.644 | 0.593 | 0.626 |
|                                                                   | taud        |        | 1.832 | 1.801 | 1.625 | 1.763 | 2.152 | 2.325 | 2.378 | 2.267 | 2.043 | 1.777 | 1.851 | 1.738 |
|                                                                   | Ebn at Noon |        | 704   | 709   | 653   | 693   | 789   | 827   | 846   | 823   | 764   | 674   | 686   | 646   |
|                                                                   | Edh at Noon |        | 198   | 215   | 264   | 232   | 155   | 129   | 123   | 138   | 171   | 217   | 193   | 213   |
| All-Sky Solar Radiation                                           | RadAvg      |        | 2.63  | 3.25  | 3.56  | 4.57  | 5.21  | 5.20  | 4.95  | 4.71  | 4.19  | 3.54  | 3.09  | 2.42  |
|                                                                   | RadStd      |        | 0.46  | 0.49  | 0.35  | 0.41  | 0.45  | 0.38  | 0.60  | 0.35  | 0.48  | 0.49  | 0.42  | 0.39  |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (5 neighbors) | +0.33 | N/A     | N/A    | +0.44   | +0.20 | +0.16 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page